

1. For each matrix equality, give the elementary coefficient matrix that enables to go from the right to the left. You can either multiply by the elementary matrix on the left or on the right but not on both sides.

(a) ($C_1 \leftrightarrow C_3$)

$$\begin{pmatrix} 1 & 2 & 4 & 3 & 9 \\ 2 & -4 & -1 & 1 & -5 \\ -1 & 1 & 3 & -6 & 3 \\ 1 & 2 & 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} & & & & \end{pmatrix} \begin{pmatrix} 4 & 2 & 1 & 3 & 9 \\ -1 & -4 & 2 & 1 & -5 \\ 3 & 1 & -1 & -6 & 3 \\ 1 & 2 & 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(b) ($L_2 \leftrightarrow L_3$)

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ -1 & -6 & 3 & 1 & 3 \\ 2 & 1 & -1 & -4 & -5 \\ 1 & 3 & 4 & 2 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 & -5 \\ -1 & -6 & 3 & 1 & 3 \\ 1 & 3 & 4 & 2 & 9 \end{pmatrix} \begin{pmatrix} & & & & \end{pmatrix}$$

(c) ($C_4 \leftarrow 5C_4$)

$$\begin{pmatrix} 1 & 1 & 1 & 10 & 2 \\ 2 & 1 & -1 & -20 & -5 \\ -1 & -6 & 3 & 5 & 3 \\ 1 & 3 & 4 & 10 & 9 \end{pmatrix} = \begin{pmatrix} & & & & \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 & -5 \\ -1 & -6 & 3 & 1 & 3 \\ 1 & 3 & 4 & 2 & 9 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(d) ($C_1 \leftarrow C_1 - C_4$)

$$\begin{pmatrix} -1 & 1 & 1 & 2 & 2 \\ 8 & 1 & -1 & -4 & -5 \\ -4 & -6 & 3 & 1 & 3 \\ -8 & 3 & 4 & 2 & 9 \end{pmatrix} = \begin{pmatrix} & & & & \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 & -5 \\ -1 & -6 & 3 & 1 & 3 \\ 1 & 3 & 4 & 2 & 9 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(e) ($L_4 \leftarrow L_4 + 2L_2$)

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 & -5 \\ -1 & -6 & 3 & 1 & 3 \\ 5 & 5 & 2 & -6 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 & -5 \\ -1 & -6 & 3 & 1 & 3 \\ 1 & 3 & 4 & 2 & 9 \end{pmatrix} \begin{pmatrix} & & & & \end{pmatrix}$$

2. Give the inverse of the following matrices. If you think the matrix is not invertible, that might be possible.

(a)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Justification not needed, but here are some possible justifications:

Justification 1: the inverse of the identity is the identity.

Justification 2: $I^2 = I$ so the the inverse of the identity is the identity.

Justification 3: If you apply the identity matrix (that is “if you change nothing”) and want to put back things to normal, you need to apply the identity matrix again (that is “change nothing” again).

(b)

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Justification not needed, but here are some possible justifications:

Justification 1: the inverse of an elementary permutation is this elementary permutation itself.

Justification 2: If P is an elementary permutation, then $P^2 = I$ so P is its own inverse.

Justification 3: If you apply “swap rows 1 and 2” and want to put back things to normal, you need to “swap rows 1 and 2” again.

(c)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{7} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Justification not needed, but here are some possible justifications:

Justification 1: the inverse of a diagonal matrix is the diagonal matrix with the inverse of the diagonal elements of the original matrix on the diagonal.

Justification 2: If you apply “scale row 2 by 7” and want to put back things to normal, you need to “scale row 2 by $\frac{1}{7}$ ”.

(d)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -4 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 4 & 0 & 1 \end{pmatrix}$$

Justification not needed, but here is a possible justification:

Justification 1: The inverse operation of $L_4 \leftarrow L_4 - 4L_2$ is $L_4 \leftarrow L_4 + 4L_2$.

(e)

$$\begin{pmatrix} 1 & 0 & 2 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}^{-1} = \text{this matrix is not invertible}$$

(f)

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}^{-1} = \text{this matrix is not square, so it is certainly not invertible}$$

3. Number of operations for this algorithm. (Only first order term, some explanations needed.)

```
def cholesky(A):
    n = A.shape[0]
    for j in range(0,n):
        for i in range(j,n):
            for k in range(0,j):
                A[i,j] = A[i,j] - A[i,k] * A[j,k]
    A[j,j] = np.sqrt(A[j,j])
    for i in range(j+1,n):
        A[i,j] = A[i,j] / A[j,j]
```

Note:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \quad \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Number of flops (floating-point operations) is

$$\left(\sum_{j=0}^{n-1} \sum_{i=j}^{n-1} \sum_{k=0}^{j-1} 2\right) + \left(\sum_{j=0}^{n-1} 1\right) + \left(\sum_{j=0}^{n-1} \sum_{i=j+1}^{n-1} 1\right)$$

(We go fast and we are just shooting for the higher order term.)

We discard the two terms on the right since they will be $\mathcal{O}(n)$ and $\mathcal{O}(n^2)$, and the leftmost one will dominate with $\mathcal{O}(n^3)$.

$$\left(\sum_{j=0}^{n-1} \sum_{i=j}^{n-1} \sum_{k=0}^{j-1} 2\right) = 2 \left(\sum_{j=0}^{n-1} \sum_{i=j}^{n-1} j\right) = 2 \left(\sum_{j=0}^{n-1} j(n-j)\right) = 2n \left(\sum_{j=0}^{n-1} j\right) - 2 \left(\sum_{j=0}^{n-1} j^2\right) = 2n \frac{n^2}{2} - 2 \frac{n^3}{3} + \mathcal{O}(n^2) = \frac{1}{3}n^3 + \mathcal{O}(n^2).$$

Answer: $\frac{1}{3}n^3 + \mathcal{O}(n^2)$.